

When Does a Legible Latent Axis Earn a Control? A Comparator-Bound Evaluation Contract for Latent-Control Interfaces

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Interfaces increasingly place controls beside named latent properties, implying that moving the control will reliably change the corresponding behavior. Readable directions and changed outputs do not establish that promise when prompting already provides a behavioral alternative. We contribute a comparator-bound evaluation contract: READ records local representational support; TRANSFER tests behavioral improvement over a declared prompt comparator; and the comparator, calibration warning, and evidence tier preserve what the result can support. We instantiate the contract with a DEV-selected prompt drawn from a preregistered bounded candidate set across Qwen and Llama, CAA and ITI, and three behavioral axes. None of the 12 model–method–axis tests qualified under the reported rule. The aggregate conceals different evidence profiles: skepticism is underpowered near the point-estimate floor, deliberation is mixed, and uncertainty is negative relative to the comparator under the frozen scorer but remains limited by a one-cell format-and-missingness recheck. A post hoc, author-proposed mapping translates these profiles into *Diagnostic-only*, *Unresolved*, *Unstable*, and *Eligible* affordance states. The result is a scoped worked method for deciding what a latent-control interface can honestly present, not a general steering verdict or a validated user-facing rubric.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Collaborative and social computing theory, concepts and paradigms*; • **Computing methodologies** → *Natural language processing*.

Additional Key Words and Phrases: latent steering, mechanistic interpretability, intelligent user interfaces, prompt comparator, evidence qualification, human–AI interaction

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1 Introduction

Alex is drafting a policy recommendation with an AI assistant and sees a slider labeled “uncertainty awareness.” Raising it appears to promise that the assistant will better distinguish what it knows from what it does not. The widget makes a behavioral claim. A readable direction or a changed completion does not show that the latent route improves the named outcome over instructions Alex could already give.

This is no longer a hypothetical interface pattern. Bo et al. expose activation steering through SELECT, CALIBRATE, and LEARN controls, while Anthropic’s Golden Gate Claude made interaction with an internally modified Claude 3 Sonnet publicly visible for 24 hours. [2, 4] Karny et al. further show that a neural-transparency interface can change reported trust and perceived helpfulness without changing several behavioral outcomes. [9] These precedents make the evidence behind an active latent control an interface question, not only a model-analysis question.

We ask: **what evidence should a latent-control interface require before a readable axis is presented as an actionable control?** We answer with a comparator-bound evaluation contract. READ records whether the label has local representational support. TRANSFER tests whether the intervention improves a declared behavioral outcome over a declared prompt comparator while satisfying an uncertainty rule, a 0.05 point-estimate floor, and coherence. The comparator, calibration warning, and evidence tier carry the result into the interface without transferring it across models, methods, axes, or tasks.

The worked application compares CAA and ITI on Qwen and Llama across deliberation, skepticism, and uncertainty awareness. The prompt comparator is selected on DEV from a preregistered bounded set of sixteen candidates; the

latent coefficient is also selected on DEV, and the frozen settings enter paired TEST comparisons. The design tests substitution—steering in place of the comparator—not prompt-plus-steer composition.

None of the twelve model–method–axis tests qualifies under the reported rule. That common procedural result does not imply one common scientific state. Skepticism is underpowered near the point-estimate floor; deliberation is mixed; and uncertainty is negative relative to the comparator under the frozen scorer, with a targeted Qwen–CAA recheck whose complete-case estimate remains negative but whose all-generation missingness bounds cross zero. The contract preserves these differences instead of converting 0/12 into a universal null or a single interface verdict.

The reported rule determines pass or no-pass. Separately, an author-proposed post hoc mapping translates evidence profiles into four affordance states: *Diagnostic-only*, *Unresolved*, *Unstable*, and *Eligible*. In the worked uncertainty example, this mapping leads a design team to withhold one active slider while retaining a read-only diagnostic and comparator view. This is a concrete design action for that record, not the state assigned to all twelve tests.

This paper makes three contributions:

1. **A comparator-bound qualification procedure for active latent controls.** It separates local READ evidence from behavioral TRANSFER evidence, reserves CONTROL for interface permission, and requires a declared alternative, outcome, uncertainty rule, point-estimate floor, coherence criterion, and evidence tier.
2. **A five-field interface record and state vocabulary.** The record preserves comparator, calibration, and scope through the proposed states *Diagnostic-only*, *Unresolved*, *Unstable*, and *Eligible*.
3. **A scoped worked application.** The reported CAA/ITI grid demonstrates how the procedure represents heterogeneous no-pass evidence profiles while preserving missingness, power, substitution, comparator, and assay-validity limits.

Together, these contributions turn a familiar interface question into a versioned decision record: evidence first, affordance second.

2 Related Work

2.1 From Legible Representations to Behavioral Actionability

Representation-engineering methods identify and intervene on directions inside language models. Representation Engineering centers population-level representations as objects for monitoring and manipulation; Activation Addition derives lightweight vectors from contrasting prompt pairs. [23, 26] CAA estimates directions from paired positive and negative examples, while ITI derives probe directions and modifies activations during generation. [12, 19] Instruction-specific vectors further show positive cases in which steering improves constraint following with and without explicit instructions. [21] Together, these methods establish that internal representations can be measured and manipulated under specified conventions. They do not by themselves determine what an interface label may promise. A direction can separate contrastive examples, move an activation, or alter tokens without reliably improving the behavioral outcome named by a control.

Recent benchmarks make this distinction empirically consequential. AxBench evaluates prompting, finetuning, and several representation-level methods on common concepts; its official Table 2 reports a mean overall steering score of 0.894 for Prompt and 0.239 for DiffMean after factor selection, distinct from the mutable repository leaderboard. [24] Basu et al. examine the gap between internal representation and actionable output correction in 400 clinical vignettes, including a safety-focused prompt-engineering condition in Arm 1; for Qwen, sensitivity changes from 0.451 under the original parser to 0.729 under the refined parser, showing that the measured gap is parser-sensitive. [3] These studies

compare channels or expose representation-to-output gaps. Our narrower target is the interface decision that follows: when may evidence about a specified latent intervention authorize an active latent control?

Internal reach and behavioral advantage are also distinct. Mishra et al. show that bounded prompting does not reproduce every residual state reached by steering, but internal non-surjectivity does not establish a superior user-visible outcome. [14] Heyman and Vandeputte show that trained steering can mimic prompting, while Sprejer et al. report capability-behavior trade-offs for feature steering relative to prompting. [7, 20] These results bound the present application in both directions: latent interventions need not collapse to prompts internally, but stronger or trained methods cannot inherit conclusions from the tested single-layer additive CAA/ITI convention.

This question cannot be answered by legibility alone. Work on metacognitive demands and human-centered transparency warns that information exposed by an AI system must be interpreted relative to a task and decision. [13, 22] We operationalize the evidence boundary as READ versus TRANSFER and reserve CONTROL for interface permission. READ remains model-, method-, axis-, layer-, and protocol-specific. TRANSFER additionally requires a declared outcome, alternative, uncertainty rule, point-estimate floor, and coherence criterion. The distinction is part of the proposed qualification procedure, not a causal theorem or a validated human decision rule.

2.2 Prompting as a Declared Behavioral Alternative

Prompting is both a practical interaction channel and a technical comparator. Prompt-derived AI instruments can reify intent as reusable direct-manipulation controls, while studies of non-expert prompt design show that constructing and evaluating prompts is itself difficult. [18, 25] AxBench directly places prompting beside representation-level methods. Bo et al. compare steering interfaces with an unscaffolded prompt baseline and separately test prompting on top of steering before their user study. [4] Basu et al. include prompt engineering within a set of mechanistic intervention arms. Together, these works rule out a broad novelty claim such as “the first comparison between prompts and steering.” They also show why a latent-control paper must state whether steering replaces, augments, or is compared with prompting.

Our comparator is deliberately described more narrowly than “the prompt a user could already have typed.” The procedure selects the prompt comparator on DEV from a preregistered bounded set of sixteen candidates.

The current application uses substitution: the latent intervention replaces the selected prompt channel. Control affordances in deployed interfaces may instead operate on top of a user’s instruction, making Bo et al.’s composition experiment an important construct boundary.

2.3 Mechanistic-Interpretability Interfaces and Human Evidence

HCI work has begun turning internal model representations into visible or manipulable interface elements. Bo et al. instantiate three steering designs—SELECT, CALIBRATE, and LEARN—and report heterogeneous preferences in an exploratory within-subject study. [4] Other systems expose feature steering, attribution-guided intervention, or embedding manipulation to people. [8, 11, 17] This literature establishes that latent representations can become interaction objects. It does not automatically establish what evidence should authorize an active latent control or keep its control affordance *Diagnostic-only*.

Public demonstrations extend this trajectory beyond research prototypes. Anthropic altered the strength of an internal Golden Gate Bridge feature in Claude 3 Sonnet and made the modified model available through `cClaude.ai` for 24 hours. [2] Bo et al. supply configurable steering interfaces; Golden Gate Claude supplies public exposure to an internally modified model. Our target is the evidence policy that follows these technical and interface precedents: what must one named control establish before it receives CONTROL permission?

Table 1. Nearest neighbors and the interface decision each leaves open.

Prior work	What it establishes	What remains for the present interface decision
Huang and Lim [8]	A lay interface for SAE feature steering and persona construction.	Whether each visible feature carries comparative behavioral evidence sufficient for an active control.
Bo et al. [4]	SELECT, CALIBRATE, and LEARN interfaces, an unscaffolded PROMPT baseline, and exploratory within-subject user evidence.	A per-affordance qualification against a declared selected comparator, with margin, coherence, warning, and evidence tier.
Karny et al. [9]	A passive neural-transparency interface with controlled human evidence and subjective/behavioral divergence.	When evidence licenses an active intervention rather than a prediction display.
Wu et al. (AxBench) [24]	A method-level benchmark comparing prompting with multiple representation-level interventions.	A widget-level decision tied to matched items, a declared margin, coherence, and scope.
Basu et al. [3]	A clinical representation-to-actionability study whose Arm 1 includes prompt engineering and whose gap is parser-sensitive.	A uniform per-affordance record that carries comparator and measurement limits into an interface state.
Riche et al.; Labarta et al.; Raval et al. [11, 17, 18]	Prompt instruments, an expert activation-steering workflow, and concept-guided latent manipulation.	A prompt-versus-latent behavioral qualification procedure for one language-model control.
Mishra et al. [14]	Internal residual-state non-surjectivity under bounded prompts.	Whether internal reach yields superior behavior under the declared outcome and comparator.
Heyman and Vandeputte [7]	Trained steering that mimics prompting.	New evidence for the trained method rather than transfer from the present naive additive setting.
Sprejer et al. [20]	Capability-behavior trade-offs for feature steering relative to prompting.	A preregistered CAA/ITI comparison on the present outcomes and an interface permission record.
Rimsky et al.; Li et al. [12, 19]	The CAA and ITI intervention families.	Whether the tested implementations exceed the declared comparator while preserving coherence.

Karny et al. provide the closest verified interface and human-evidence neighbor. Their neural-transparency system projects a system prompt onto persona trait vectors and presents the predictions through an interactive sunburst before users chat with their configured model. In a controlled between-subjects study, the visualization increased endpoint trust and was rated helpful, but it did not produce significant differences in prompt iteration, resulting persona scores, message engagement, or confidence in predicting unintended behavior. [9] They study passive prediction and explicitly leave active steering interfaces to future work. Their subjective/behavioral divergence also cautions against treating perceived usefulness as evidence that an interface improves actionability judgments.

Documentation frameworks such as model cards and datasheets demonstrate how structured records can bind claims to provenance, evaluation, and scope. [6, 15] Internal algorithmic-audit frameworks add a process-oriented precedent: staged development decisions produce linked records across the system lifecycle. [16] Our five-field record applies a related accountability move at the control-affordance level: it records READ, TRANSFER, comparator, calibration warning, and evidence tier for one proposed active latent control. These precedents motivate traceability; the contribution is a qualification procedure and worked application rather than a documentation standard.

2.4 Positioning Summary

Together, prior work establishes the ingredients of the problem: methods can expose and manipulate representations; prompting is a viable behavioral channel; and latent evidence can become an interface object. What remains open is how one proposed control affordance should carry its comparator, measurement warnings, and scope into a design decision. We address that step with a control-affordance qualification record. The contribution is the scoped translation from evidence to control affordance, not a claim that one rule is universally necessary or sufficient.

3 Method

3.1 Evidence Status and Study Overview

Exact values below trace to the committed result artifacts, preregistrations, analysis code, evidence ledger, and paper manifests in the repository. The core grid is stored under `results/arm_full/`; READ artifacts are stored under `results/gpu_7b_2026-07-23/c1/` and `results/llama_c1_facade_2026-07-24/`; and E-0011, E-0013, E-0014, and E-0015 retain their scoped roles in the evidence ledger. Generated paper tables and figures are rebuilt from these committed artifacts without manually editing their values.

The method has two levels. First, a control-affordance qualification record specifies what evidence a proposed active latent control would need to claim actionability. Second, a computational application fills the record for CAA and ITI relative to a DEV-selected prompt comparator drawn from a preregistered bounded candidate set. The application crosses two model families, two steering methods, and three behavioral TRANSFER axes. READ is evaluated separately with exploratory CAA-style directions and includes a fourth, focus-related axis that is not part of the twelve-test TRANSFER denominator.

3.2 Five-Field Qualification Record

The record contains five fields.

1. **READ** specifies the model, method, direction construction, layer, task, and measurement under which an internal axis is legible. READ is a diagnostic precondition, not behavioral evidence.
2. **TRANSFER** records the behavioral outcome, intervention, estimand, uncertainty, point-estimate floor, and coherence result.
3. **COMPARATOR** identifies the declared behavioral alternative, candidate set, and selection procedure. In this application it is a DEV-selected prompt drawn from a preregistered bounded set of sixteen candidates; the field preserves the provenance limits detailed below.
4. **CALIBRATION WARNING** states which contrast is being interpreted—for example, steer versus prompt rather than steer versus an unsteered baseline—and records missingness or scoring threats that can reverse the reading.
5. **EVIDENCE TIER** binds the record to a model, method, axis, task, layer, protocol, and evidence status so that a result cannot silently transfer to another setting.

The proposed vocabulary maps each record to exactly one of four formal affordance states. *Diagnostic-only* retains local READ evidence for inspection but grants no CONTROL permission because TRANSFER is untested or clearly misses the interval or point-estimate criterion. *Unresolved* marks unsupported READ, inadequate power, missing data, or an interval spanning zero. *Unstable* marks a coherence failure. *Eligible* requires every declared behavioral criterion to pass within the evidence tier. These states are author-proposed: no blinded independent analyst has reproduced field completion or state assignment, and no inter-rater validation is planned.

3.3 Models, Methods, and Behavioral Axes

The worked application uses Qwen2.5-7B-Instruct and Meta-Llama-3-8B-Instruct, with the latter loaded from the NousResearch identical-weights mirror after gated access blocked the canonical repository. Both run in CUDA float16. The committed artifacts record the selected hidden-state layers by deliberation/skepticism/uncertainty: Qwen-CAA uses 20/20/20, Qwen-ITI uses 20/19/17, Llama-CAA uses 12/14/10, and Llama-ITI uses 8/14/11. The intervention

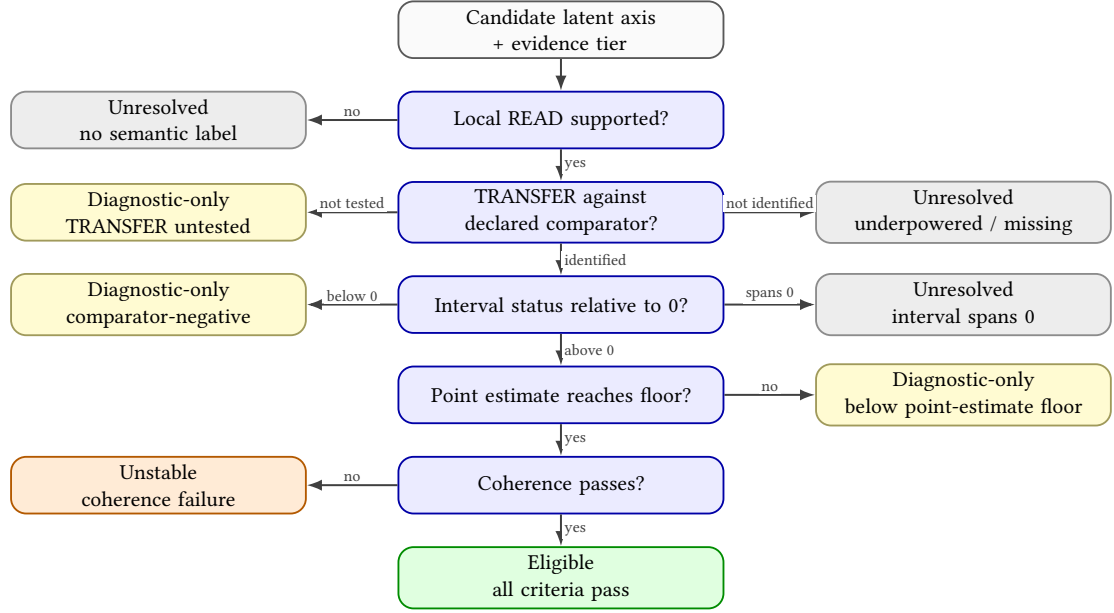


Fig. 1. Proposed evidence-to-affordance flow. Color distinguishes decisions (blue) from the four formal states: Diagnostic-only (yellow), Unresolved (gray), Unstable (orange), and Eligible (green). Text labels carry the same distinctions. Outputs retain comparator, warning, and evidence tier.

hooks the output of decoder block $L - 1$, which corresponds to Hugging Face hidden state L , and adds the direction at every token position during prefill and generation. Activation extraction renders one user turn through the model chat template and pools the last non-padding token. Immutable upstream model revision hashes were not recorded.

CAA uses a positive-minus-negative mean-difference direction that is unit-normalized before injection. ITI fits a logistic probe to the same 28-pair extraction split, selects a nondegenerate layer no shallower than 20% of model depth, unit-normalizes the probe direction, and multiplies it by the frozen projected-activation sample standard deviation. Both methods select a coefficient from $\{2, 4, 6, 8, 12, 16, 24\}$ on DEV, but equal coefficients do not imply equal injected norms: ITI injects $\alpha\sigma_L$, so no common norm bound of 24 applies.

The reported CAA convention is

$$\begin{aligned} v_a &= \mathbb{E}_{x \in A_a^+} [h_L(x)] - \mathbb{E}_{x \in A_a^-} [h_L(x)], \\ \hat{v}_a &= v_a / \|v_a\|, \quad h'_L = h_L + \alpha \hat{v}_a. \end{aligned} \quad (1)$$

The common additive notation preserves method-specific scale:

$$h'_L = h_L + \alpha s_{m,a,L} \hat{u}_{m,a,L}, \quad s_{CAA,a,L} = 1, \quad s_{ITI,a,L} = \sigma_{a,L}. \quad (2)$$

Equations 1 and 2 make the method-specific scaling explicit; they do not supply the missing extraction, layer, or hook provenance.

The three TRANSFER axes use task accuracy for deliberation, false-premise rejection for skepticism, and 1 – Brier for uncertainty awareness. These operational outcomes do not exhaust the psychological constructs named

by the axes. For uncertainty, both conditions request an answer and confidence. The deterministic parser accepts Confidence:/conf forms, interprets values above one as percentages, clamps them to $[0, 1]$, and imputes missing confidence as 0.5, which makes $1 - \text{Brier} = 0.75$ regardless of correctness. The scorer and parser are implemented in `src/cognitive_console/eval/scorers.py` and exercised by the repository test suite.

3.4 DEV-Selected Prompt Comparator and Substitution Design

The preregistered bounded candidate set contains sixteen instructions with varied wording and specificity. The reported procedure selects the prompt comparator from that set on DEV, while the available provenance does not establish TEST blindness, authorship independence, budget parity, fairness, or global optimality. The DEV-selected prompt comparator is therefore a declared study-specific alternative rather than an estimate of the best prompt a user or optimizer could produce.

The core experiment is a substitution design. The prompt condition uses the selected instruction. The steering condition uses a neutral carrier instruction plus the latent intervention. This isolates the channels under the reported convention but does not test how an active control behaves on top of a user’s prompt. Prompt-plus-steer evidence is pending, so no composition claim enters the paper.

3.5 Generations, Outcomes, and Paired Estimand

For each item and condition, the setup draws five sampled generations at temperature 0.7 with a 64-new-token cap and batch size 16. The frozen item pools contain 60 deliberation, 60 skepticism, and 80 uncertainty items; seed 20260723 yields DEV/TEST counts of 20/40, 20/40, and 27/53. Outcomes are averaged over generations within item before computing paired steer-minus-prompt differences on TEST. No item is removed after the fixed split: parse failures score through the deterministic outcome rules, including the 0.5 confidence imputation, rather than being excluded. The token cap may constrain deliberation, particularly multi-step reasoning.

For item i , axis a , condition s , and generation j , the reported paired estimand is

$$\bar{y}_{ia}^{(s)} = \frac{1}{k} \sum_{j=1}^k y_{iaj}^{(s)}, \quad d_{ia} = \bar{y}_{ia}^{(\text{steer})} - \bar{y}_{ia}^{(\text{prompt})}. \quad (3)$$

Equation 3 makes item-level pairing explicit: the three TEST denominators are 40, 40, and 53 items, each carrying five generations, with no performance-based exclusions.

The reported uncertainty analysis resamples TEST items rather than individual generations using 10,000 item-cluster bootstrap replicates. A within-cell Bonferroni correction across three axes produces two-sided 98.33% intervals; it does not control familywise error over the full descriptive twelve-test grid.

3.6 Reported Qualification Rule

For axis a , let $\bar{d}_a$ be the paired mean difference between steering and the selected prompt, I_a its item-cluster bootstrap interval, g_a^S the steering degradation score, and g_a^0 its baseline counterpart. The reported rule qualifies an axis only when

$$\text{qualify}(a) \iff 0 \notin I_a \wedge \bar{d}_a \geq \delta \wedge g_a^S \leq \tau g_a^0 + \epsilon. \quad (4)$$

with $\delta = 0.05$, $\tau = 1.5$, and $\epsilon = 0.02$. Equation 4 requires an interval excluding zero and a point estimate at least δ ; it does **not** require the interval’s lower bound to exceed δ . The protocol and executable adjudicator were frozen on 2026-07-23. The degradation score is one minus the distinct trigram ratio for each generation; scores are averaged within the steering and unsteered cells, and the steering mean must satisfy $g_a^S \leq 1.5g_a^0 + 0.02$. The implementation adds 10^{-12} only as floating-point comparison tolerance.

The legacy procedure further labels a model–method cell positive when at least two axes pass, conditional when one passes, and negative when none pass. This cell taxonomy is distinct from the proposed interface-state vocabulary: an underpowered no-pass maps to *Unresolved*, not automatically to *Diagnostic-only* or *Unstable*. No independent state assignment is available to test that translation.

3.7 Exploratory READ Measurement

READ uses contrastive CAA-style directions in both model families. The facade compares strongest-prompt displacement from a neutral origin with same-origin positive-pole reach and requires the ratio and its 95% bootstrap interval to lie below one. Each model contributes one exploratory run. Qwen ratios are 0.583 [0.488, 0.680] for deliberation, 0.548 [0.434, 0.670] for skepticism, 0.713 [0.517, 0.910] for uncertainty, and 2.844 for focus (overshoot). Llama ratios are 0.872 [0.798, 0.937], 0.628 [0.581, 0.673], 1.000 [0.839, 1.147], and 0.471 [0.306, 0.630], respectively. Thus deliberation and skepticism satisfy the criterion in both models, uncertainty only in Qwen, and focus only in Llama. READ results remain exploratory, CAA-specific, and non-transferable to ITI.

The reported same-origin facade ratio is

$$r_a = \frac{\langle \bar{h}_{\text{prompt},a} - \bar{h}_{\text{neutral},a}, \hat{v}_a \rangle}{\langle \bar{h}_{\text{extract},+,a} - \bar{h}_{\text{neutral},a}, \hat{v}_a \rangle}. \quad (5)$$

Equation 5 is a same-origin projection ratio, not behavioral evidence or an ITI READ test.

3.8 Manipulation and Measurement Checks

The reported diagnostic suite separates outcome-pipeline sensitivity, endpoint liveness, injection-scale adequacy, and internal propagation. It perturbs uncertainty with a random direction, compares a direct refusal prompt with bounded unit-direction CAA, contrasts natural and injected CAA norms, sweeps raw-magnitude Qwen–CAA coefficients, and inspects refusal-leading token mass. Movement at one link does not establish target-specific latent transfer or validate the end-to-end assay.

The first-token logit diagnostic is explicitly non-confirmatory because refusal-leading tokens can also begin ordinary or hedged answers. Its values are excluded from claim support and reported only as propagation context.

4 Results

4.1 One Headline, Three Evidence Profiles

The qualification procedure produces one headline but three different evidence profiles. Across the Qwen and Llama model families, two methods (CAA and ITI), and three axes, 0 of 12 model–method–axis tests qualified under the reported rule; because no latent behavioral positive control passed, this is a scoped procedural classification under unresolved assay validity, not a general transfer diagnosis. Skepticism calls for more resolution, deliberation resists a single superiority-or-equivalence summary, and uncertainty reaches a sharper comparator-relative result whose scope is limited by format and missingness. The Results follow those profiles rather than treating the grid as a uniform null.

Table 2. Paired TEST steer-minus-prompt outcomes across CAA/ITI $\times$ Qwen/Llama, with Bonferroni two-sided confidence intervals at level 0.98333 and meaningful margin $\delta = 0.05$.

Method	Model	Axis	N_{dev}	N_{test}	α	Δ [CI]	Gate / pass
CAA	Qwen2.5-7B	Deliberation	20	40	2	+0.015 [-0.040, +0.070]	ok / fail
CAA	Qwen2.5-7B	Skepticism	20	40	6	-0.080 [-0.225, +0.045]	ok / fail
CAA	Qwen2.5-7B	Uncertainty	27	53	8	-0.228 [-0.370, -0.092]	ok / fail
CAA	Llama-3-8B	Deliberation	20	40	4	+0.025 [-0.030, +0.075]	ok / fail
CAA	Llama-3-8B	Skepticism	20	40	2	+0.000 [-0.190, +0.200]	ok / fail
CAA	Llama-3-8B	Uncertainty	27	53	8	-0.072 [-0.103, -0.034]	ok / fail
ITI	Qwen2.5-7B	Deliberation	20	40	4	+0.020 [+0.000, +0.055]	ok / fail
ITI	Qwen2.5-7B	Skepticism	20	40	2	-0.100 [-0.270, +0.060]	ok / fail
ITI	Qwen2.5-7B	Uncertainty	27	53	12	-0.103 [-0.136, -0.069]	ok / fail
ITI	Llama-3-8B	Deliberation	20	40	2	+0.015 [-0.040, +0.060]	ok / fail
ITI	Llama-3-8B	Skepticism	20	40	2	+0.000 [-0.195, +0.205]	ok / fail
ITI	Llama-3-8B	Uncertainty	27	53	6	-0.084 [-0.115, -0.049]	ok / fail

Notes. Pass/fail is computed from the unrounded frozen results. Displayed means and intervals are rounded to three decimals; a fail can therefore have a small positive displayed estimate or boundary. A pass requires an unrounded CI excluding 0, mean $\geq \delta = 0.05$, and the coherence gate to pass, so ITI $\times$ Qwen deliberation remains fail because its mean is below δ . The supplementary artifact record provides the DEV-selected prompt identifier and text for each axis and model.

4.2 READ Is Model-, Axis-, and Method-Specific

The exploratory READ facade reports a three-of-four aggregate pattern in each model, but the composition differs. Deliberation and skepticism satisfy the reported facade criterion in both model families; uncertainty does so only in Qwen, while focus does so only in Llama and overshoots in Qwen. These results come from one exploratory run per model using CAA-style directions. They do not establish run stability, a model-independent semantic axis, or corresponding READ evidence for ITI.

This heterogeneity matters for the evidence record. An aggregate “three of four” label would hide which affordance is legible in which model, and a CAA READ result would silently grant evidence to a different construction method. We therefore keep READ tied to its model, direction family, axis, and measurement rather than treating it as an intrinsic property of an axis name.

4.3 TRANSFER No-Pass States Differ by Axis

Table 2 exposes the complete four-cell grid rather than asking the aggregate 0/12 label to carry the evidence. Every model–method–axis test is a procedural no-pass under the reported rule, but the table does not convert an underpowered or mixed comparison into a resolved failure or a withheld-control conclusion.

Skepticism calls for more resolution, not an ineffectiveness label. Minimum detectable positive effects (MDEs) were 0.188, 0.268, 0.225, and 0.279 across the four model–method cells, compared with a declared point-estimate floor of 0.05. The study could not resolve effects near its own floor, so the next action is more information rather than an ineffectiveness claim.

Deliberation resists both superiority and equivalence summaries. Two ITI cells fell inside a ± 0.05 equivalence interval in a post-hoc TOST, while the CAA cells remained underpowered. The evidence is mixed: the exploratory analysis bounds two cells but cannot become a preregistered equivalence result.

Uncertainty is comparator-negative under the frozen scorer, but not grid-wide resolved. All four steer-minus-prompt contrasts were negative, but only one supplied format-and-missingness recheck retains a negative complete-case contrast, its all-generation bounds cross zero, and three cells did not receive this recheck. This result supports a comparator-relative warning with an explicit measurement boundary.

Table 3. No-pass evidence profiles differ by behavioral axis and imply different next actions.

Axis	Reported evidence profile	Proposed affordance state (reason)
Skepticism	MDEs far exceed the 0.05 point-estimate floor	Unresolved (underpowered)
Deliberation	Small mixed differences; two ITI cells described as equivalent post hoc; CAA cells underpowered	Unresolved (mixed; no preregistered equivalence)
Uncertainty	Frozen steer-minus-prompt contrasts negative; one-cell complete-case recheck remains negative but all-generation missingness bounds cross zero	Unresolved (missingness; three cells unrechecked)

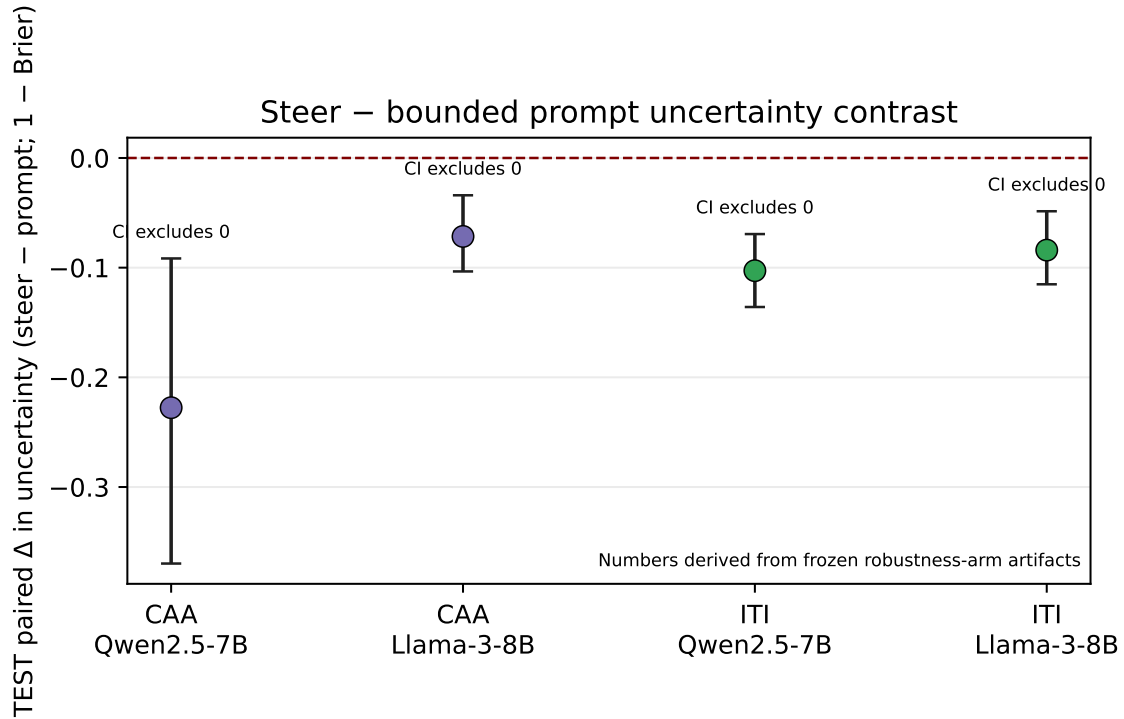


Fig. 2. Paired steer-minus-bounded-prompt uncertainty effects and Bonferroni-corrected intervals across the four tested model-method cells.

4.4 Comparator Choice Changes the Calibration Reading

In the Qwen-CAA format-and-missingness recheck, steering remained close to the unsteered baseline—about +0.011 in parseable-confidence rate and +0.0008 in 1 – Brier—while the comparator scored higher under the reported outcome. The three-way comparison therefore supports a comparator-relative warning, not a claim that steering directly harmed baseline calibration.

The format-and-missingness recheck exposes a reporting confound: the parseable-confidence rate was 0.826 for steering and 0.445 for the prompt condition. Among complete pairs, steer minus prompt was -0.337 with a 95% confidence interval of $[-0.508, -0.165]$, but this analysis conditions on post-generation parseability, its adversarial

Table 4. The calibration interpretation changes with comparator and missingness scope in the Qwen-CAA format-and-missingness recheck.

Quantity	Reported value	Defensible interpretation
Steer minus unsteered compliance	about +0.011	Steering is near the unsteered baseline in this lineage.
Steer minus unsteered 1 – Brier	about +0.0008	The steer-versus-prompt gap cannot be described as direct baseline harm.
Parseable-confidence rate	steer 0.826; prompt 0.445	Conditions differ in confidence reporting.
Complete-case steer minus prompt	−0.337 [−0.508, −0.165]	Negative only among pairs where both outputs are complete.
All-generation adversarial bound	[−0.478, +0.250]	Sign is not identified under unrestricted missingness.
Other uncertainty cells	not rechecked	No grid-wide format/missingness conclusion.

all-generation bound widens to $[-0.478, +0.250]$ and crosses zero, and the other three uncertainty cells did not receive this recheck. The sign is therefore not identified under unrestricted missingness, and no grid-wide resolved uncertainty effect follows.

4.5 Split Sensitivity Does Not Supply New-Item Replication

The reported no-pass verdict recurred over five split seeds. The original split had already been observed before four additional splits were prospectively frozen, and all splits reused the same item pool. This result reduces concern that the verdict is peculiar to one DEV/TEST partition. It does not test new items, independent replication, or external validity.

4.6 Manipulation Checks Do Not Validate the Assay

Three checks narrow the explanation space. **Can the endpoint move?** A direct refusal prompt reaches 95% refusal. **Can the outcome pipeline register perturbation?** A random uncertainty direction changes the measured outcome by -0.24 , with a 95% confidence interval of $[-0.36, -0.12]$. **Can a latent intervention that satisfies the coherence criterion supply a known behavioral positive?** No tested latent control passes. The checks establish endpoint and measurement responsiveness while leaving target-specific latent manipulation validity unresolved.

Scale and propagation checks sharpen that boundary rather than closing it. The natural refusal class-difference norm was about 216 versus an injected norm of at most 24 under bounded CAA; a raw-magnitude Qwen-CAA sweep still produced no qualifying pass in the nondegenerate tested range, and its highest-scale point was degenerate. A non-confirmatory token-mass diagnostic moved strongly without scored refusals and remains excluded from claim support.

4.7 Mechanism Remains Open

The preregistered whitened-Mahalanobis off-manifold test failed its criterion in all four cells. This is a mechanism null, not support for a competing mechanism. The present evidence cannot distinguish whether no-pass classifications arise from direction construction, intervention scale, layer/hook choice, model adaptation, outcome sensitivity, generation constraints, or a true absence of useful behavioral transfer.

Table 5. Manipulation and measurement checks narrow the explanation space without resolving target-specific assay sensitivity.

Check	Reported result	What it establishes / does not establish
Random uncertainty direction	$-0.24 [-0.36, -0.12]$	Outcome pipeline registers perturbation; not target-specific validation.
Direct refusal prompt	95.5% refusal	Endpoint is responsive to prompting; not a latent positive control.
Bounded unit-direction CAA	0% refusal for all tested coefficients	Exposes scale/implementation concern; does not validate transfer.
Natural versus injected CAA norm	about 216 versus at most 24	Shows mismatch for bounded CAA; not a common ITI norm bound.
Raw-magnitude Qwen–CAA	no qualifying pass in nondegenerate tested range	Weakens simple under-scaling; assay validity remains open.
First-token logit proxy	about +13.9 nats, 0/5 scored refusals	Non-confirmatory propagation context; excluded from claims.

Table 6. Worked five-field record for Qwen–CAA uncertainty under the current evidence.

Field	Worked entry
READ	Exploratory CAA-style READ satisfies the reported facade criterion for uncertainty on Qwen; one run, no ITI transfer.
TRANSFER COMPARATOR	No-pass relative to the DEV-selected prompt comparator; not evidence that latent steering generally fails. DEV-selected prompt from a preregistered bounded set of sixteen candidates; provenance limits follow the Method definition.
CALIBRATION WARNING	Steering is near the unsteered baseline in the one rechecked lineage; complete-case steer-minus-prompt remains negative, but adversarial all-generation missingness bounds cross zero and three cells are unchecked.
EVIDENCE TIER	Qwen2.5-7B-Instruct, CAA single-layer additive intervention at hidden-state layer 20 (decoder-block-19 output hook at every token position), uncertainty task/outcome, seed 20260723, reported substitution protocol.

The procedure therefore returns more than pass or no-pass: under the proposed mapping, it records a next diagnostic action for this worked application. Underpowered evidence requests more data, format-sensitive evidence requests measurement repair, and READ-only evidence permits inspection without granting CONTROL permission.

5 Interface-Facing Qualification Record

5.1 From Evaluation Output to Affordance State

The record prevents three substitutions in interface reasoning: READ for TRANSFER, absolute output change for comparison against the declared alternative, and evidence from one tier for another. It binds those obligations to one proposed state: *Diagnostic-only*, *Unresolved*, *Unstable*, or *Eligible*. The mapping is applied here as a worked author synthesis; its validation boundary is consolidated in Limitations.

5.2 Worked Record: Qwen–CAA Uncertainty

Under the proposed mapping, this record is *Unresolved*: the TRANSFER test is no-pass relative to the comparator and the all-generation contrast remains partially unidentified. Local READ evidence can still support inspection. “Unresolved under this record” is narrower than “uncertainty cannot be controlled.”

COMPARATOR-BOUND CONTROL RECORD

Artifact-derived interface states; no user-effect claim.

DELIBERATION UNRESOLVED	UNCERTAINTY-AWARENESS WITHHELD CONTROL
READ HOLDS; local CAA evidence only. TRANSFER Mixed resolution: ITI exploratory equivalence; CAA underpowered. BOUNDED PROMPT COMPARATOR TEST prompt mean=0.020 EXACT EVIDENCE TIER Frozen 2x2 grid; mixed resolution; split-sensitivity only. NEXT ACTION More evidence at the registered margin or a new powered evaluation.	READ HOLDS; local CAA evidence only. TRANSFER FAIL; 4/4 frozen contrasts comparator-negative. DIRECT VS BASELINE Near-baseline only in the CAA x Qwen recheck. FORMAT RECHECK Complete-case result only for CAA x Qwen. MISSINGNESS Adversarial bounds cross zero. UNRECHECKED SCOPE Other 3 method-model cells unchecked. EXACT EVIDENCE TIER Frozen 2x2 comparator result; targeted CAA x Qwen recheck.

Fig. 3. Artifact-derived cards make the comparator, evidence profile, tier, and next design action visible.

5.3 Illustrative Design Scenario

Return to Alex’s policy-writing assistant. The prototype uncertainty slider has local Qwen–CAA READ support, but its TRANSFER record is no-pass relative to the comparator. The targeted recheck shows steering near the unsteered baseline, a negative complete-case steer-minus-prompt estimate, and missingness bounds that cross zero.

For this record, the team withholds the active slider and keeps a read-only inspection trace. The comparator view explains what the prompt channel achieved; the warning distinguishes steer-versus-prompt from steer-versus-baseline; and the evidence tier keeps the Qwen–CAA result from migrating to ITI or Llama. Withholding the slider is the design action in this scenario, not a uniform classification of every no-pass test.

5.4 Display Requirements Implied by the Record

An interface that exposes this record would need to keep the relative comparison visible. Showing only the negative steer-minus-prompt estimate could imply that steering harmed baseline calibration; the Qwen–CAA format-and-missingness recheck instead places steering near baseline and the comparator above it. The record must also expose missingness rather than letting a complete-case interval stand in for all generations. Finally, the evidence tier must prevent Qwen–CAA READ from appearing as evidence for Llama, ITI, another layer, or another task.

Human-AI guidelines support communicating capabilities and limitations, but interpretability tools can also be misunderstood or misused. [1, 10] The record therefore exposes the reason for a state rather than presenting a green or red badge. Whether people understand or benefit from this presentation remains a separate empirical question addressed in Limitations.

6 Discussion

6.1 What Remains Useful after No-Pass

A readable direction can still support inspection, debugging, hypothesis formation, or a read-only trace. The worked READ results make that value concrete: deliberation and skepticism satisfy the facade criterion in both model families, while uncertainty and focus differ by model. The alternative is not “ship a slider” or “hide the representation.” It is to preserve local evidence without turning visibility into a behavioral promise.

The three evidence profiles point to different actions. Skepticism requests more resolution; deliberation calls for a better-targeted test; uncertainty calls for measurement repair and a comparator-specific warning. This translation from evidence profile to interface action is the record’s practical contribution.

6.2 Why the Comparator Changes the Interface Claim

An active latent control makes a claim about consequences, not merely movement. In the Qwen–CAA format-and-missingness recheck, the three-way relation—steering near baseline, the comparator higher, and confidence output incomplete—changes the message from “steering harms calibration” to “steering does not match this comparator under this measurement.” The comparator field determines what the contrast means.

The procedure asks designers to disclose the comparator rather than canonize it. In this application, the comparator is the DEV-selected prompt defined in Method. The procedure also distinguishes substitution from deployment: the present experiment replaces that prompt with steering, whereas a user-facing slider normally acts on top of an instruction.

6.3 Qualification Is a Versioned Lifecycle

A new model, method, direction, comparator, item pool, or interface version creates a new evidence tier rather than inheriting the current result. This lifecycle keeps the artifact useful without treating a local classification as permanent or universal.

7 Limitations, Ethics, and Future Work

7.1 Evidence Provenance and Reproducibility

The repository contains the core result directories, frozen preregistrations, executable adjudicator, evidence ledger, paper manifests, generated tables and figures, and independent audit summaries. The committed JSON artifacts can regenerate the claim-bearing paper tables and figures without GPU access. The author-proposed affordance-state assignments remain non-generated judgments and have not been independently reproduced.

The available lineage records exact layers, hook convention, contrast splits, prompt wrappers, generation defaults, TEST denominators, split seeds, and degradation scorer. Remaining reproducibility limits include immutable upstream model revision hashes and incomplete environment/data hashes in the older E-0006 registry entries; reproducing raw generations therefore requires reconstructing those model snapshots and software environments, whereas regenerating the paper artifacts uses the committed result JSON and manifests.

7.2 Assay Validity and Statistical Resolution

Because no latent behavioral positive control passes, the no-pass grid cannot distinguish a true absence of actionability from limitations in direction construction, scale, hook/layer selection, normalization, generation, outcome, or scoring. Prompt endpoint liveness and random-direction sensitivity show that parts of the pipeline respond; they do not validate

the target intervention. The raw-magnitude follow-up narrows simple under-scaling only for Qwen single-layer CAA, and CAA/ITI coefficients do not imply a common injected norm.

Resolution also differs by claim. READ is exploratory, single-run, and CAA-specific. Skepticism is underpowered relative to the point-estimate floor, deliberation’s equivalence descriptions are post hoc, and uncertainty remains format- and missingness-sensitive. Five split seeds over one item pool address partition sensitivity rather than replication, and within-cell correction does not establish a familywise global null across the grid.

7.3 Comparator, Substitution, and Interface Construct

The prompt comparator is selected on DEV from a preregistered bounded candidate set. Its authorship independence, TEST blindness, and resource parity are not established; a different candidate set or search budget could change the contrast. Conclusions therefore remain relative to this declared alternative.

The application substitutes steering for the selected prompt, whereas a product control would ordinarily act in the presence of a user’s instruction. Prompt-plus-steer evidence is pending, so the paper makes no composition claim. The four-state mapping was synthesized from the observed evidence profiles rather than frozen in advance, and no independent analyst reproduced its assignments.

7.4 Human Evidence, Ethics, and Future Validation

The current study contains no human result, so the paper does not claim that people understand the record, calibrate reliance, or make better control-granting decisions. Any future human evaluation requires the applicable ethics determination, consent and recruitment records, compensation, exclusion flow, and privacy handling.

A structured record can itself create false assurance when its comparator is narrow, its measurement incomplete, or its assignment author-dependent. Its warnings and failure states should therefore remain visible rather than becoming a compliance badge. The most direct validation priorities are a known-positive latent intervention under the same pipeline and independent application of a frozen state rubric; new comparators, item pools, or method families would each require re-evaluation. Human studies should distinguish comprehension, decision calibration, behavior, and perceived helpfulness because subjective approval alone cannot establish actionability. [5, 9]

8 Conclusion

A latent control should not be licensed by readability or output movement alone. The comparator-bound contract asks whether the intervention improves the named behavior over a declared alternative, then preserves READ, TRANSFER, comparator, calibration warning, and evidence tier in the interface decision.

In the tested Qwen/Llama and CAA/ITI setting, none of the twelve tests qualifies under the reported rule. The useful result is not a universal null: skepticism is underpowered, deliberation is mixed, and uncertainty is comparator-negative under the frozen scorer but limited by missingness. The contract keeps those profiles distinct. In the worked uncertainty scenario, that distinction supports a concrete action—retain diagnostic evidence, expose the comparator and warning, and withhold one unsupported slider. Evidence determines the affordance, and every new model, method, or task must earn its own record.

A Author-Proposed Assignment Checklist

The checklist makes the proposed procedure inspectable; it does not supply independent or inter-rater validation.

1. **Specify candidate and tier.** Record model, method, direction construction, layer, task, version, and protocol.
2. **Define outcome and READ evidence.** State what behavior the proposed control names and whether local representational evidence supports the label.
3. **Declare the behavioral alternative.** Record comparator set, selection procedure, budget, and known provenance limits without calling it fair or optimal.
4. **Run TRANSFER and coherence.** Report paired estimand, uncertainty rule, point-estimate floor, missingness, exclusions, and coherence result.
5. **Issue a provisional state.** Assign exactly one formal state—*Diagnostic-only*, *Unresolved*, *Unstable*, or *Eligible*—and attach its reason, comparator, warning, and evidence tier.

Any model, method, axis, layer, task, comparator, outcome, or protocol change creates a new record.

B Non-Claim-Bearing Logit Diagnostic

This diagnostic is qualitative context and is excluded from claim support. It uses five probes per axis and measures first-position log-sum-exp changes for refusal-leading tokens, including tokens such as “I” that may begin ordinary or hedged answers. At coherent $\beta = 1$, the refusal direction raises this token mass by about 13.9 nats on average while 0 of 5 probes match the refusal markers. One probe rises by about 16.9 nats with a byte-identical greedy completion; at $\beta = 2$, generation becomes repetitive. Token-proxy movement is not scored refusal and does not establish a working latent handle.

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